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Lab Overview and Objectives

This laboratory work exercises end-to-end version control workflows using Git and GitHub. The primary focus is mastering multi-branch development models, isolating feature updates, managing pull requests, conducting interactive code reviews with reviewer comments, resolving merge workflows, and cleaning up local and remote branches. Furthermore, team collaboration was simulated and verified.

Task 1: Repository Initialization and Local Setup

Objective and Details: Initializing the local Git repository, configuring user identity, SSH authentication, initial file tracking, and establishing link with GitHub remote repository.

Commands Executed:

```
git --version
ssh-keygen -t ed25519 -C "marihamalik02@gmail.com"
cat ~/.ssh/id_ed25519.pub
git config --global user.name "Mariha Malik"
git config --global user.email "marihamalik02@gmail.com"
git config --list
ls -a .git
git init
echo "#Lab02 Git Practice" >> README.md
git add README.md
git commit -m "Initial commit"
git remote add origin git@github.com:mareeh58/DevOps_Lab2_Practice.git
git push -u origin main -f
```

Verification Evidence (Screenshots):



```
mareeh58@ubuntu:~$ git clone git@github.com:mareeh58/DevOps_Lab2_practice.git
Cloning into 'DevOps_Lab2_practice'...
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Compressing objects: 100% (2/2), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (3/3), done.
mareeh58@ubuntu:~$ git config --global user.name "Mariha Malik"
mareeh58@ubuntu:~$ git cpnfig --global user.email "marihamalik02@gmail.com"
git: 'cpnfig' is not a git command. See 'git --help'.

The most similar command is
config
mareeh58@ubuntu:~$ git config --global user.email "marihamalik02@gmail.com"
mareeh58@ubuntu:~$
```

git_identity

```

The most similar command is
config
mareeh58@ubuntu:~$git config --global user.email "marihamalik02@gmail.com"
mareeh58@ubuntu:~$git config --list
user.name=Mariha Malik
user.email=marihamalik02@gmail.com
mareeh58@ubuntu:~$

```

git_config_list

```

config
mareeh58@ubuntu:~$git config --global user.email "marihamalik02@gmail.com"
mareeh58@ubuntu:~$git config --list
user.name=Mariha Malik
user.email=marihamalik02@gmail.com
mareeh58@ubuntu:~$cd DevOps_Lab2_practice
mareeh58@ubuntu:~/DevOps_Lab2_practice$ls -la .git
.. HEAD      description  index  logs      packed-refs
.. config  hooks      info    objects  refs
mareeh58@ubuntu:~/DevOps_Lab2_practice$

```

git_folder

```

mareeh58@ubuntu:~$git config --list
user.name=Mariha Malik
user.email=marihamalik02@gmail.com
mareeh58@ubuntu:~$cd DevOps_Lab2_practice
mareeh58@ubuntu:~/DevOps_Lab2_practice$ls -la .git
.. HEAD      description  index  logs      packed-refs
.. config  hooks      info    objects  refs
mareeh58@ubuntu:~/DevOps_Lab2_practice$rm -rf .git
mareeh58@ubuntu:~/DevOps_Lab2_practice$git status
fatal: not a git repository (or any of the parent directories): .git
mareeh58@ubuntu:~/DevOps_Lab2_practice$

```

delete_git

```

fatal: not a git repository (or any of the parent directories): .git
mareeh58@ubuntu:~/DevOps_Lab2_practice$git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: will change to 'main' in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this warning
hint: call:
hint:   git config --global init.defaultBranch <name>
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:   git branch -m <name>
hint: Disable this message with 'git config set advice.defaultBranchName false'
Initialized empty Git repository in /home/marihamalik/DevOps_Lab2_practice/.git/
mareeh58@ubuntu:~/DevOps_Lab2_practice$

```

git_init

```
mareeh58@ubuntu:~/DevOps_Lab2_practice$ echo "#Lab02 Git Practice" >> README.md
mareeh58@ubuntu:~/DevOps_Lab2_practice$ git add README.md
mareeh58@ubuntu:~/DevOps_Lab2_practice$ git commit -m "Initial commit"
Command 'git' not found, but can be installed with:
sudo apt install git
mareeh58@ubuntu:~/DevOps_Lab2_practice$ git commit -m "Initial commit"
[main (root-commit) 07c0640] Initial commit
1 file changed, 3 insertions(+)
create mode 100644 README.md
mareeh58@ubuntu:~/DevOps_Lab2_practice$
```

first_commit

```
mareeh58@ubuntu:~/DevOps_Lab2_practice$ git remote add origin git@github.com:mareeh58/DevOps_Lab2_Practice.git
mareeh58@ubuntu:~/DevOps_Lab2_practice$ git push -u origin main -f
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 283 bytes | 283.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:mareeh58/DevOps_Lab2_Practice.git
+ f45b37a... 07c0640 main -> main (forced update)
branch 'main' set up to track 'origin/main'.
mareeh58@ubuntu:~/DevOps_Lab2_practice$
```

first_push

```
branch 'main' set up to track 'origin/main'.
mareeh58@ubuntu:~/DevOps_Lab2_practice$ echo "Adding some test notes" >> notes.txt
mareeh58@ubuntu:~/DevOps_Lab2_practice$ git status
On branch main
Your branch is up to date with 'origin/main'.

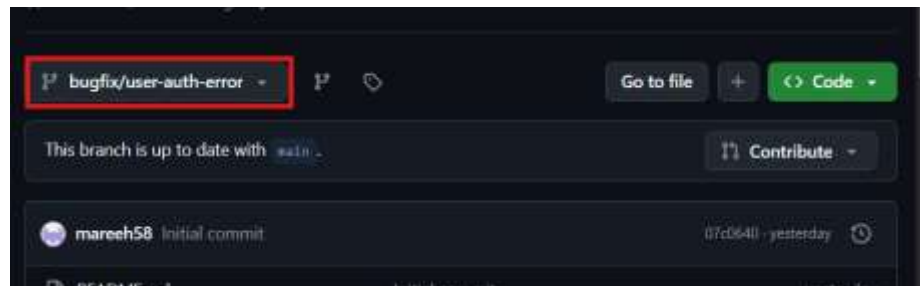
Untracked files:
  (use "git add <file>..." to include in what will be committed)
  notes.txt

nothing added to commit but untracked files present (use "git add" to track)
mareeh58@ubuntu:~/DevOps_Lab2_practice$
```

status1

```
mareeh58@ubuntu:~/DevOps_Lab2_practice$ git add notes.txt
mareeh58@ubuntu:~/DevOps_Lab2_practice$ git commit -m "Adding notes.txt"
[main 3af862f] Adding notes.txt
1 file changed, 1 insertion(+)
create mode 100644 notes.txt
mareeh58@ubuntu:~/DevOps_Lab2_practice$
```

commit_notes



bugfix_branch_gui

```
mareeh58@ubuntu:~/DevOps_Lab2_practice$git fetch origin
From github.com:mareeh58/DevOps_Lab2_Practice
* [new branch]      bugfix/user-auth-error -> origin/bugfix/user-auth-error
mareeh58@ubuntu:~/DevOps_Lab2_practice$git checkout bugfix/user-auth-error
branch 'bugfix/user-auth-error' set up to track 'origin/bugfix/user-auth-error'.
Switched to a new branch 'bugfix/user-auth-error'
mareeh58@ubuntu:~/DevOps_Lab2_practice$git branch
* bugfix/user-auth-error
  main
mareeh58@ubuntu:~/DevOps_Lab2_practice$
```

bugfix_branch_local

```
mareeh58@ubuntu:~/DevOps_Lab2_practice$git checkout -b feature/db-connection
Switched to a new branch 'feature/db-connection'
mareeh58@ubuntu:~/DevOps_Lab2_practice$git checkout -b feature/db-connection
Switched to a new branch 'feature/db-connection'
mareeh58@ubuntu:~/DevOps_Lab2_practice$git push origin feature/db-connection
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'feature/db-connection' on GitHub by visiting:
remote:   https://github.com/mareeh58/DevOps_Lab2_Practice/pull/new/feature/d
remote: b-connection
remote:
To github.com:mareeh58/DevOps_Lab2_Practice.git
* [new branch]      feature/db-connection -> feature/db-connection
mareeh58@ubuntu:~/DevOps_Lab2_practice$
```

feature_db_branch

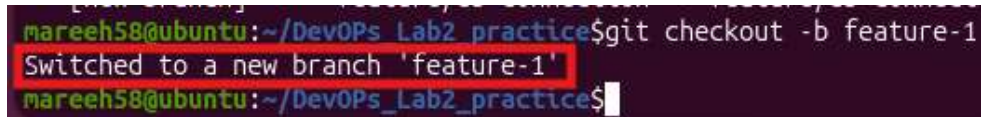
Task 2: Creating and Switching Between Branches

Objective and Details: Demonstrating branch isolation by creating dedicated feature branches for independent task execution.

Commands Executed:

```
git checkout -b feature-1  
git status
```

Verification Evidence (Screenshots):



```
mareeh58@ubuntu:~/DevOPs_Lab2_practice$git checkout -b feature-1  
Switched to a new branch 'feature-1'  
mareeh58@ubuntu:~/DevOPs_Lab2_practice$
```

branch_create

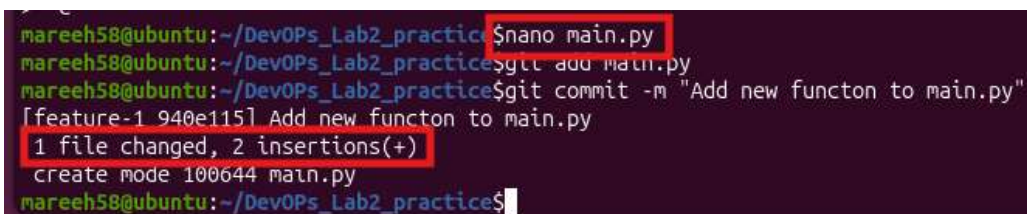
Task 3: Making Changes on a Feature Branch

Objective and Details: Modifying codebase files on the feature branch without affecting the production 'main' branch.

Commands Executed:

```
nano main.py  
git add main.py  
git commit -m "Add new function to main.py"
```

Verification Evidence (Screenshots):



```
mareeh58@ubuntu:~/DevOPs_Lab2_practice$nano main.py  
mareeh58@ubuntu:~/DevOPs_Lab2_practice$git add main.py  
mareeh58@ubuntu:~/DevOPs_Lab2_practice$git commit -m "Add new function to main.py"  
[feature-1 940e115] Add new function to main.py  
1 file changed, 2 insertions(+)  
create mode 100644 main.py  
mareeh58@ubuntu:~/DevOPs_Lab2_practice$
```

feature_commit

Task 4: Pushing Feature Branch to Remote GitHub

Objective and Details: Publishing the local feature branch to GitHub to make changes available for team collaboration.

Commands Executed:

```
git push origin feature-1
```

Verification Evidence (Screenshots):

```

create mode 100644 main.py
mareeh58@ubuntu:~/DevOps_Lab2_practice$git push origin main
Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Delta compression using up to 2 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (8/8), 773 bytes | 193.00 KiB/s, done.
Total 8 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), done.
To github.com:mareeh58/DevOps_Lab2_Practice.git
   07c0640..cd0529b  main -> main
mareeh58@ubuntu:~/DevOps_Lab2_practice$git push origin feature-1
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'feature-1' on GitHub by visiting:
remote:   https://github.com/mareeh58/DevOps_Lab2_Practice/pull/new/feature-1
remote:
To github.com:mareeh58/DevOps_Lab2_Practice.git
 * [new branch]      feature-1 -> feature-1
mareeh58@ubuntu:~/DevOps_Lab2_practice$

```

push_branches

Task 5: Merging Feature Branch Locally

Objective and Details: Switching back to the primary branch and merging changes made in feature-1 into main.

Commands Executed:

```
git checkout main
git merge feature-1
```

Verification Evidence (Screenshots):

```

mareeh58@ubuntu:~/DevOps_Lab2_practice$git checkout main
Switched to branch 'main'
Your branch is ahead of 'origin/main' by 1 commit.
(use "git push" to publish your local commits)
mareeh58@ubuntu:~/DevOps_Lab2_practice$git merge feature-1
Merge made by the 'ort' strategy.
 main.py | 2 ++
1 file changed, 2 insertions(+)
create mode 100644 main.py
mareeh58@ubuntu:~/DevOps_Lab2_practice$

```

merge

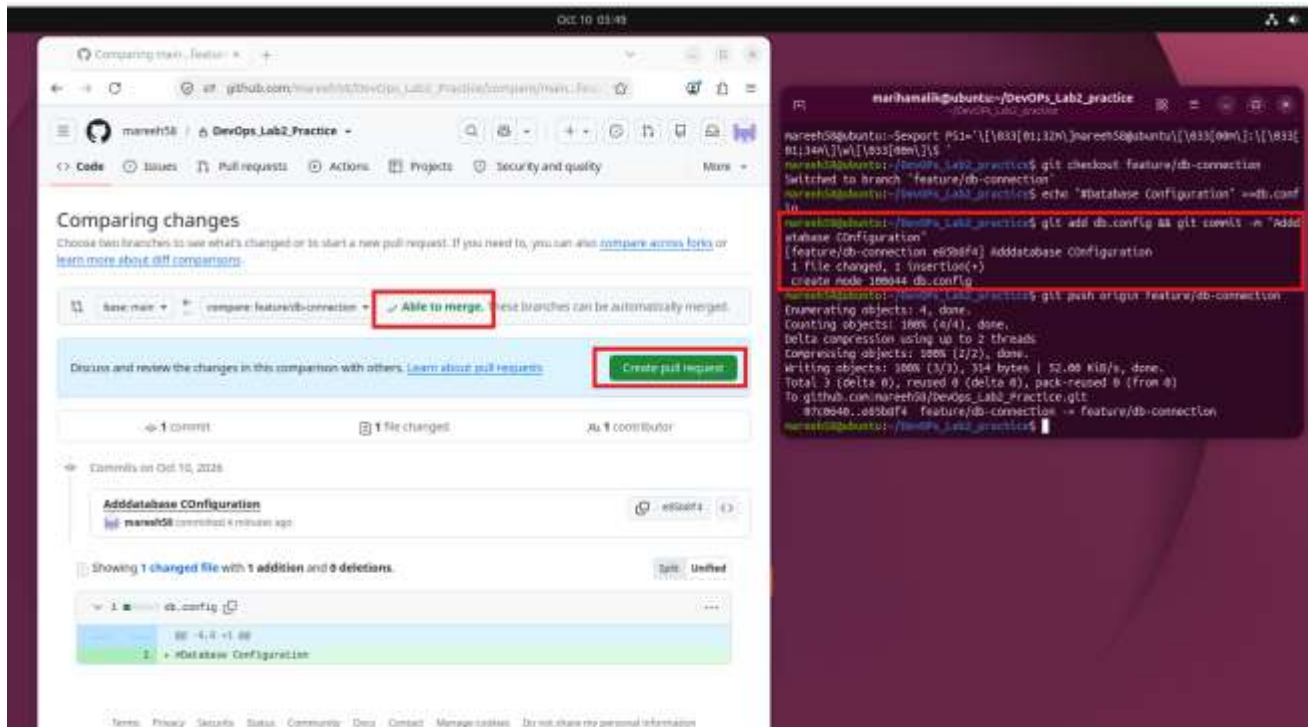
Tasks 8 - 10: Pull Request Creation, Merging & Remote Branch Deletion

Objective and Details: Navigating to GitHub UI, opening a Pull Request for feature/db-connection into main, reviewing changes, merging the PR, and deleting the branch.

Commands Executed:

```
# Performed via GitHub UI & synchronized locally
git checkout main
git pull origin main
```

Verification Evidence (Screenshots):



pr_creation

Browser tabs: Adddatabase COnfiguration x +

Address bar: github.com/mareeh58/DevOps_Lab2_Practice/pull/1

Navigation: <> Code Issues **Pull requests** Actions Projects Security and quality More

Adddatabase COnfiguration #1

Merged mareeh58 merged 1 commit into `main` from `feature/db-connection`  now

Conversation 0 Commits 1 Checks 0 Files changed 1 +1

 **mareeh58** commented 2 minutes ago ...

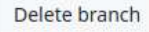
No description provided.

  Adddatabase COnfiguration e85b8f4

  **mareeh58** merged commit 73ddc7f into `main` now 

Pull request successfully merged and closed

You're all set — the `feature/db-connection` branch can be safely deleted.



Reviewers: No reviews. Still in progress? [Convert to draft](#)

Assignees: No one—[assign yourself](#)

Labels: None yet

Projects: None yet

Milestone: No milestone

Development

github.com/mareeh58/DevOps_Lab2_Practice/issues

pr_merge

Oct 10 05:55

Addatabase Configurati x +

github.com/mareeh58/DevOps_Lab2_Practice/pull/1

Merged Addatabase Configuration #1
mareeh58 merged 1 commit into `main` from `feature/db-connection`

Commit history:

- Addatabase Configuration e85b8f4
- mareeh58 merged commit 73ddc7f into `main` 2 minutes ago
- mareeh58 deleted the `feature/db-connection` branch [now](#) [Restore branch](#)

Pull request successfully merged and closed
You're all set — the branch has been merged.

Add a comment

Write Preview H B I

Add your comment here...

Markdown is supported Paste, drop, or click to add files

[Remember, contributions to this repository should follow our GitHub Community Guidelines.](#)

ProTip! Add comments to specific lines under [Files changed](#).

Labels None yet

Projects None yet

Milestone No milestone

Development Successfully merging this pull request may close these issues. None yet

Notifications Customize
[Unsubscribe](#)
You're receiving notifications because you modified the open/close state.

1 participant

[Lock conversation](#)
[Archive pull request](#)

branch_delete

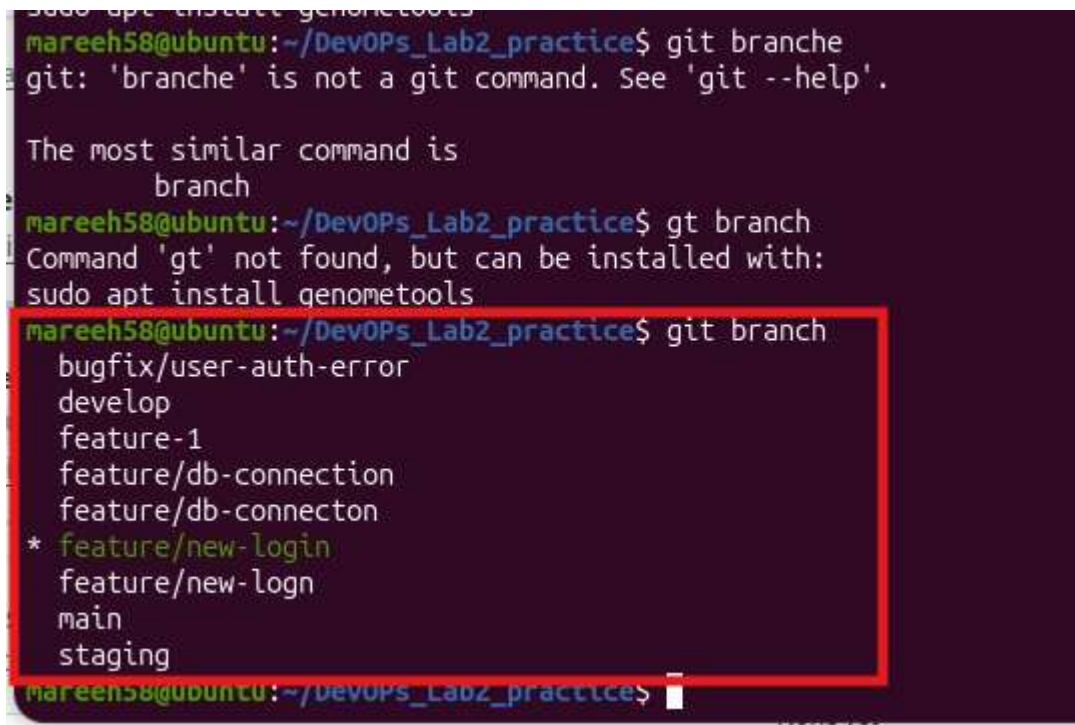
Task 11: Detailed Branch Strategy (Develop / Staging / Production)

Objective and Details: Simulating enterprise branching strategy: feature branches merge into 'develop', 'develop' merges into 'staging' for testing, and 'staging' merges into 'main' for production release.

Commands Executed:

```
git branch
git checkout develop
git merge feature/new-login
git checkout staging
git merge develop
git checkout main
git merge staging
git push origin main develop staging
```

Verification Evidence (Screenshots):



The screenshot shows a terminal window with the following commands and output:

```
mareeh58@ubuntu:~/DevOps_Lab2_practice$ git branche
git: 'branche' is not a git command. See 'git --help'.

The most similar command is
    branch
mareeh58@ubuntu:~/DevOps_Lab2_practice$ gt branch
Command 'gt' not found, but can be installed with:
sudo apt install genomertools
mareeh58@ubuntu:~/DevOps_Lab2_practice$ git branch
bugfix/user-auth-error
develop
feature-1
feature/db-connection
feature/db-connecton
* feature/new-login
feature/new-logn
main
staging
mareeh58@ubuntu:~/DevOps_Lab2_practice$
```

The output of the `git branch` command is highlighted with a red box, showing the current branch `feature/new-login` with an asterisk.

branch_strategy



```
staging
mareeh58@ubuntu:~/DevOps_Lab2_practice$ git checkout develop
Switched to branch 'develop'
mareeh58@ubuntu:~/DevOps_Lab2_practice$ git merge feature/new-login
Updating 73ddc7f..1145d58
Fast-forward
 login.py | 1 +
 1 file changed, 1 insertion(+)
 create mode 100644 login.py
mareeh58@ubuntu:~/DevOps_Lab2_practice$ git checkout staging
Switched to branch 'staging'
mareeh58@ubuntu:~/DevOps_Lab2_practice$ git merge develop
Updating 73ddc7f..1145d58
Fast-forward
 login.py | 1 +
 1 file changed, 1 insertion(+)
 create mode 100644 login.py
mareeh58@ubuntu:~/DevOps_Lab2_practice$
```

branch_merges

```
mareeh58@ubuntu:~/DevOps_Lab2_practice$ git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
mareeh58@ubuntu:~/DevOps_Lab2_practice$ git merge staging
Updating 73ddc7f..1145d58
Fast-forward
 login.py | 1 +
 1 file changed, 1 insertion(+)
 create mode 100644 login.py
mareeh58@ubuntu:~/DevOps_Lab2_practice$ git push origin main develop staging
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 304 bytes | 30.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:mareeh58/DevOps_Lab2_Practice.git
 73ddc7f..1145d58  main -> main
* [new branch]      develop -> develop
* [new branch]      staging -> staging
mareeh58@ubuntu:~/DevOps_Lab2_practice$
```

final_merge

Task 12: Code Review Workflow (PR Review, Comments, Rejection & Update)

Objective and Details: Simulating full code review lifecycle on PR #2: setting base/compare branches, creating title/description, assigning reviewer/assignee, adding inline code review comment, rejecting/closing PR, pushing author updates, and final merge.

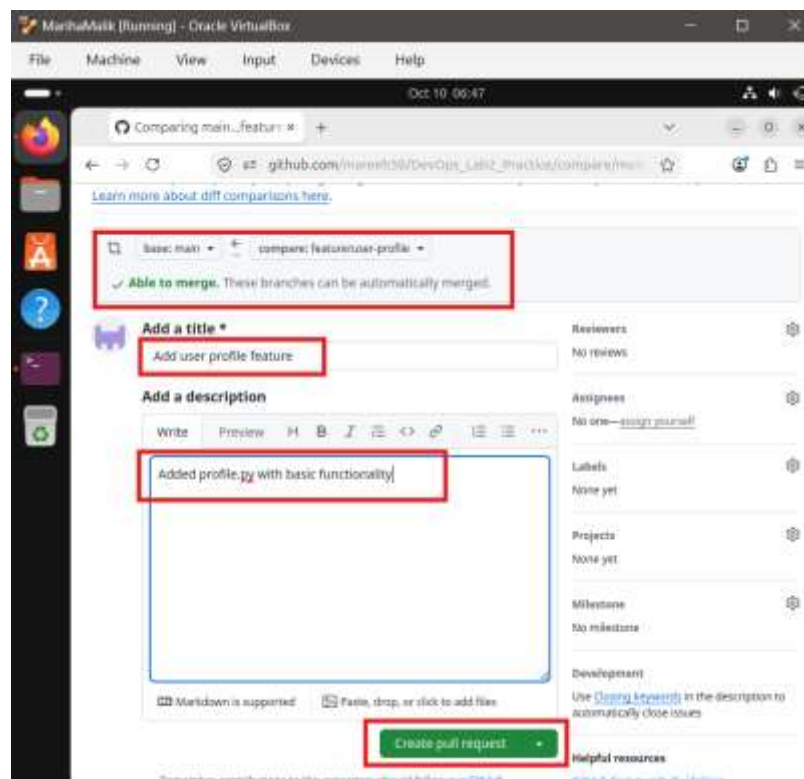
Commands Executed:

```
git checkout -b feature/user-profile  
echo "print('user profile module')" > profile.py  
git commit -am "add profile feature initial code"  
git push -u origin feature/user-profile
```

Code update addressing review:

```
echo "print('Profile updated with proper error handling')" >> profile.py  
git commit -am "Fix: Add error handling as requested in review"  
git push origin feature/user-profile
```

Verification Evidence (Screenshots):



pr_create_details

Oct 10 06:53

Add user profile feature | x

github.com/mareeh58/DevOps_Lab2_Practice/pull/2

mareeh58 / DevOps_Lab2_Practice

<> Code Issues Pull requests 1 Actions Projects More

Ready to merge Code

Add user profile feature #2

Open mareeh58 wants to merge 1 commit into main from feature/user-profile

Conversation 0 Commits 1 Checks 0 Files changed 1 +1

mareeh58 commented 4 minutes ago

Added profile.py with basic functionality

add profile feature initial code 3239fe8

mareeh58 self-assigned this 1 minute ago

No conflicts with base branch

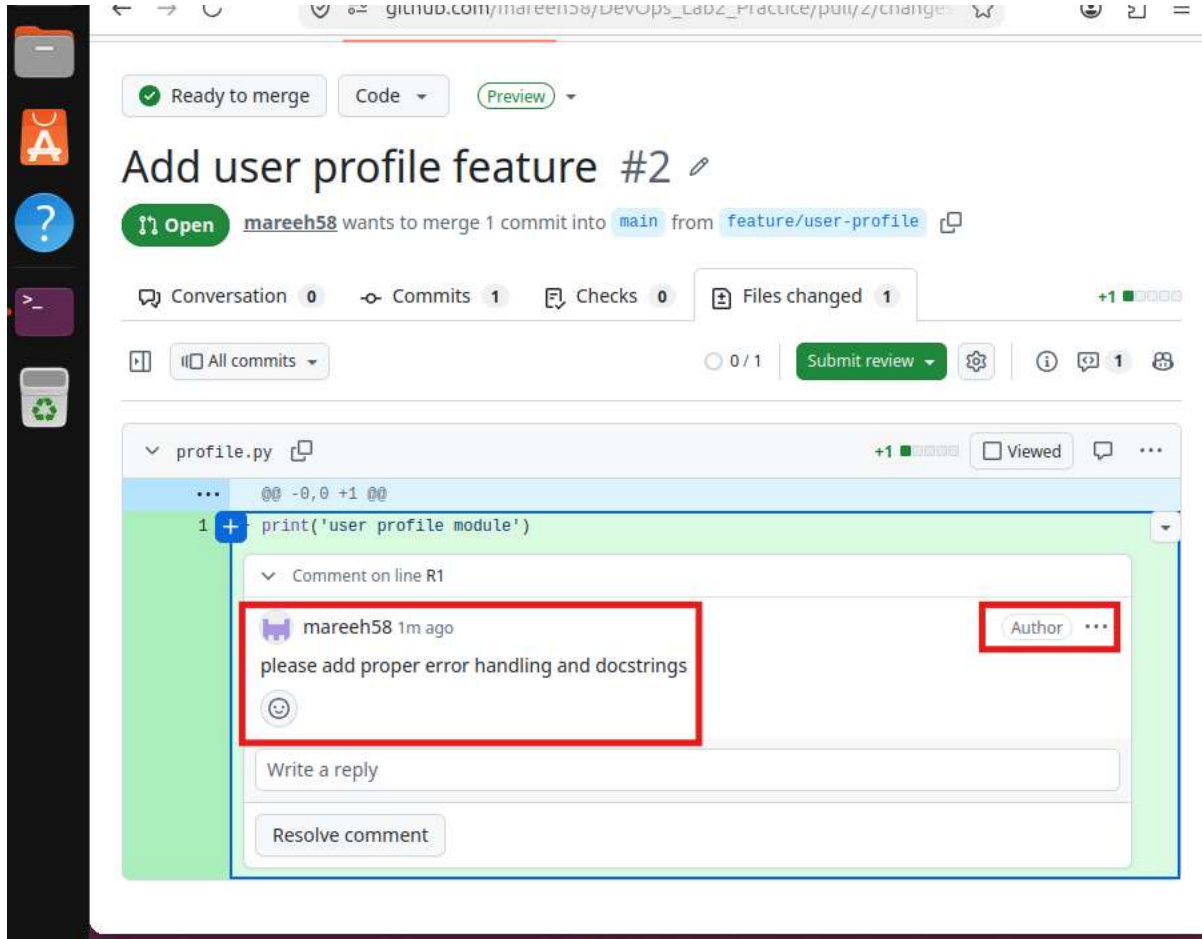
Reviewers
No reviews
Still in progress? [Convert to draft](#)

Assignees
mareeh58

Labels
None yet

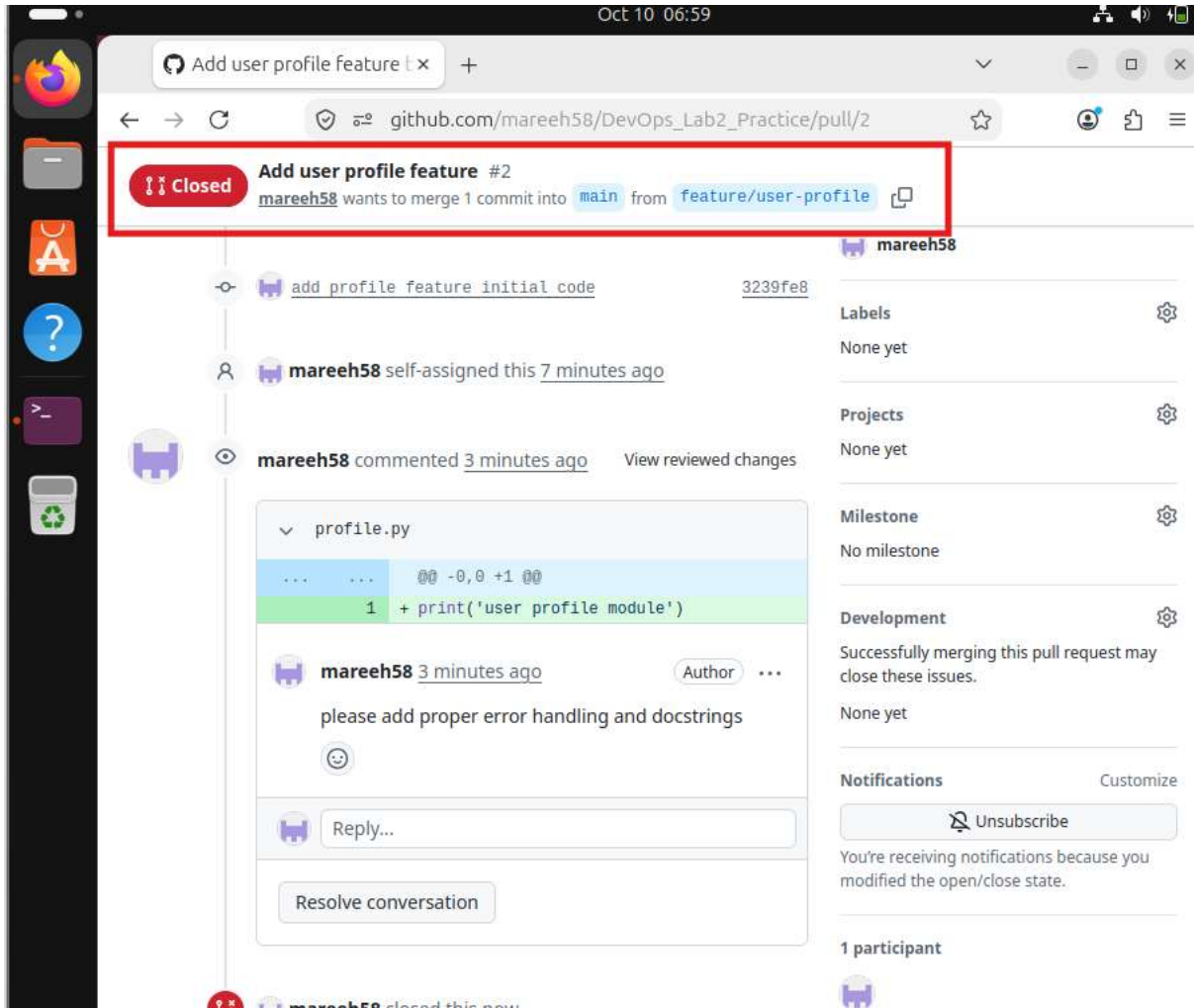
Projects
None yet

pr_assigned_reviewer



The screenshot shows a GitHub pull request page. At the top, the title is "Add user profile feature #2". Below the title, it says "mareeh58 wants to merge 1 commit into main from feature/user-profile". The interface includes tabs for "Ready to merge", "Code", and "Preview". There are statistics for "Conversation 0", "Commits 1", "Checks 0", and "Files changed 1". A "Submit review" button is visible. The main content area shows a diff for "profile.py" with a single line added: `print('user profile module')`. Below the code, a comment from "mareeh58" 1m ago says "please add proper error handling and docstrings". The comment is highlighted with a red box, and the "Author" field is also highlighted with a red box. There is a "Write a reply" input field and a "Resolve comment" button.

pr_request_changes



Oct 10 06:59

Add user profile feature #2

mareeh58 wants to merge 1 commit into main from feature/user-profile

Labels: None yet

Projects: None yet

Milestone: No milestone

Development: Successfully merging this pull request may close these issues. None yet

Notifications: Unsubscribe

You're receiving notifications because you modified the open/close state.

1 participant

add profile feature initial code 3239fe8

mareeh58 self-assigned this 7 minutes ago

mareeh58 commented 3 minutes ago View reviewed changes

```
profile.py
...    ...    @@ -0,0 +1 @@
1 + print('user profile module')
```

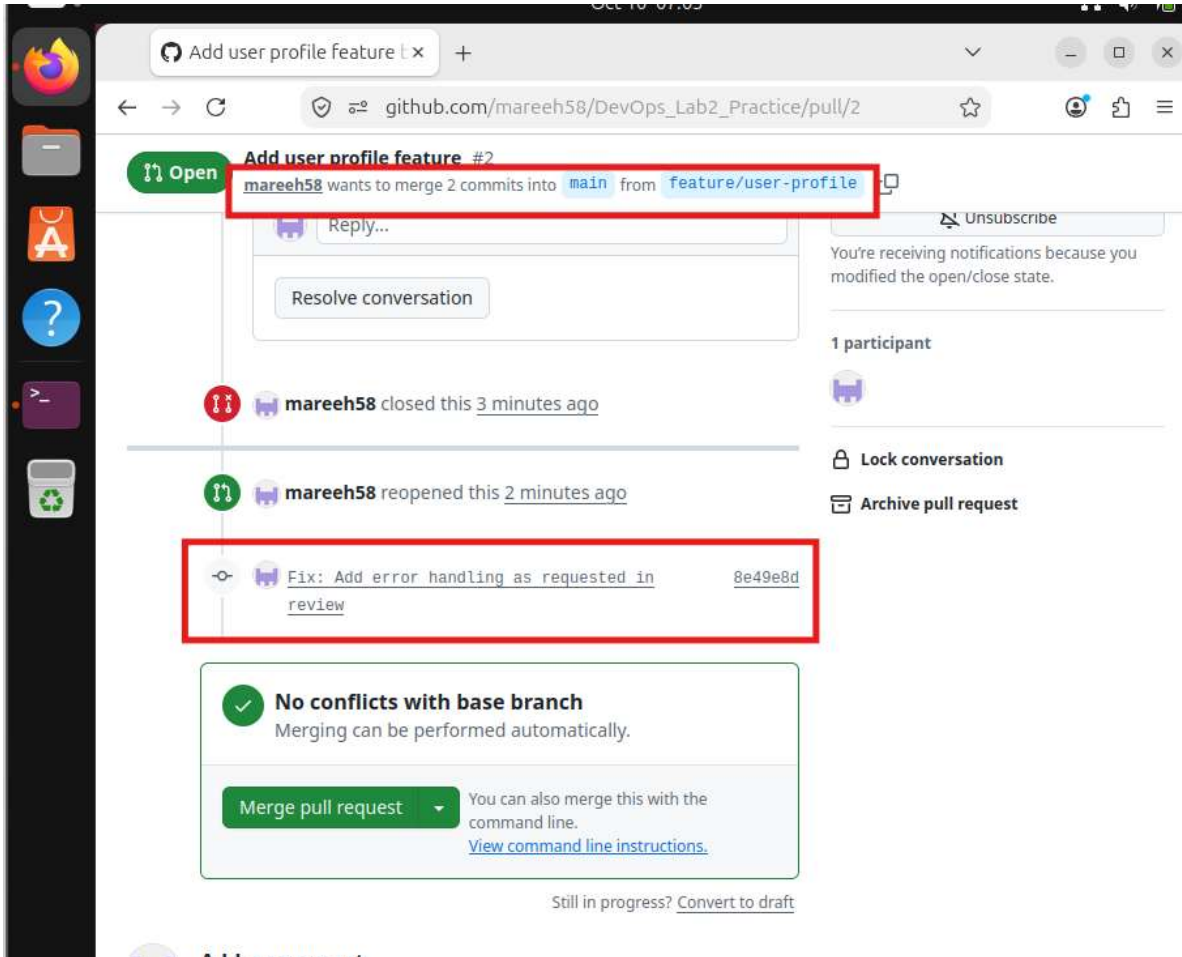
mareeh58 3 minutes ago Author ...

please add proper error handling and docstrings

Reply...

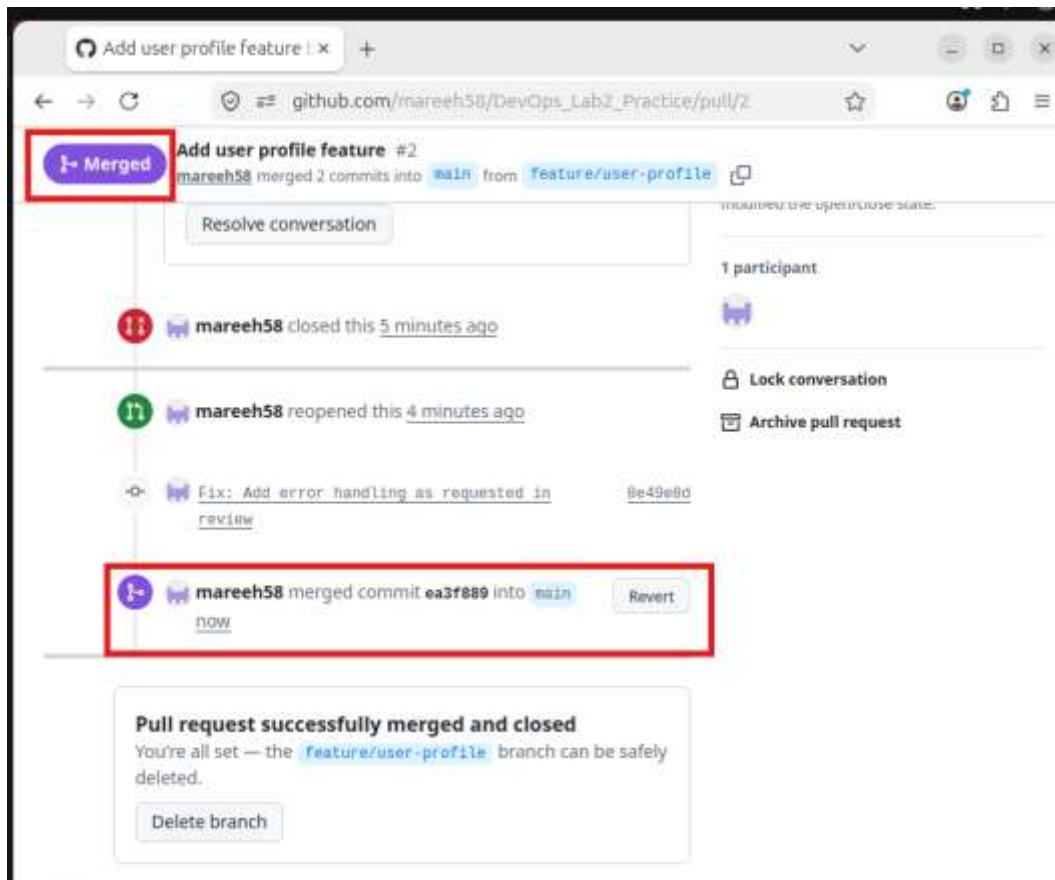
Resolve conversation

pr_rejected

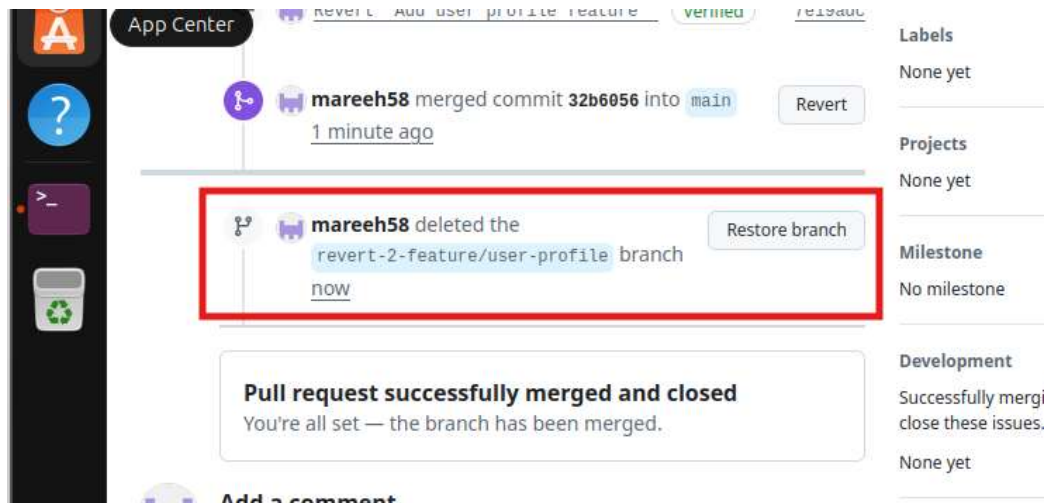


The screenshot shows a web browser window displaying a GitHub pull request. The browser's address bar shows the URL `github.com/mareeh58/DevOps_Lab2_Practice/pull/2`. The pull request title is "Add user profile feature #2". A red box highlights the header area where it says "mareeh58 wants to merge 2 commits into main from feature/user-profile". Below this, there are status updates: "mareeh58 closed this 3 minutes ago" and "mareeh58 reopened this 2 minutes ago". Another red box highlights a comment that says "Fix: Add error handling as requested in review" with a commit hash "8e49e8d". At the bottom, a green box indicates "No conflicts with base branch" and provides a "Merge pull request" button. The right sidebar contains options like "Unsubscribe", "Lock conversation", and "Archive pull request".

pr_updated_with_commits



pr_merged



pr_branch_deleted

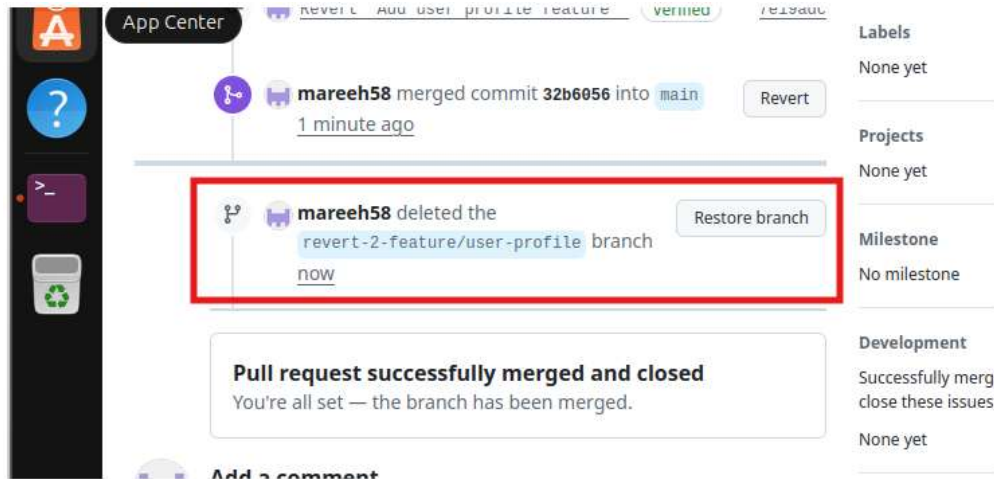
Task 13: Branch Cleanup Best Practices

Objective and Details: Pruning obsolete feature branches from both remote (GitHub) and local environments after successful integration.

Commands Executed:

```
git push origin --delete feature-1
git checkout main
git pull origin main
git branch -d feature-1
git branch -d feature/user-profile
git branch
```

Verification Evidence (Screenshots):



remote_branch_deleted

```
mareeh58@ubuntu:~/DevOps_Lab2_practice$ git push origin --delete feature/cleanup
- demo
To github.com:mareeh58/DevOps_Lab2_Practice.git
- [deleted]          feature/cleanup-demo
mareeh58@ubuntu:~/DevOps_Lab2_practice$
```

remote_branch_delete_cmd

Bonus Task: Simulated Team Collaboration

Objective and Details: Simulating two contributors (Mariha Malik & Fatima) working on the collab branch, committing separate author identities into notes.txt, pushing to GitHub, adding collaborator invitation, and merging into main.

Commands Executed:

```
git checkout -b collab
echo "User 1 (Mariha): ..." >> notes.txt
git commit -m "Add user1 fun fact"
echo "User 2 (Fatima): ..." >> notes.txt
git commit --author="Fatima <fatima@example.com>" -m "Add User 2 fun fact"
git push -u origin collab
```

```
git log --oneline
git checkout main
git merge collab
git push origin main
```

Verification Evidence (Screenshots):

```
tes.txt
mareeh58@ubuntu:~/DevOps_Lab2_practice$ git add notes.txt
mareeh58@ubuntu:~/DevOps_Lab2_practice$ git commit -m "Add user1 fun fact"
[collab b60372c] Add user1 fun fact
1 file changed, 1 insertion(+)
mareeh58@ubuntu:~/DevOps_Lab2_practice$ echo "User 2 (Fatima): Enjoys automation and scripting!" >>notes.txt
mareeh58@ubuntu:~/DevOps_Lab2_practice$ git add notes.txt
mareeh58@ubuntu:~/DevOps_Lab2_practice$ git commit --author="Fatima <fatima@example.com>" -m "Add User 2 fun fact"
[collab d8c0755] Add User 2 fun fact
Author: Fatima <fatima@example.com>
1 file changed, 1 insertion(+)
mareeh58@ubuntu:~/DevOps_Lab2_practice$ git push -u origin collab
Enumerating objects: 8, done.
Counting objects: 100% (8/8), done.
Delta compression using up to 2 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (6/6), 620 bytes | 620.00 KiB/s, done.
Total 6 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (3/3), completed with 1 local object.
remote:
remote: Create a pull request for 'collab' on GitHub by visiting:
remote:   https://github.com/mareeh58/DevOps_Lab2_Practice/pull/new/collab
remote:
To github.com:mareeh58/DevOps_Lab2_Practice.git
* [new branch]      collab -> collab
branch 'collab' set up to track 'origin/collab'.
mareeh58@ubuntu:~/DevOps_Lab2_practice$ git log --oneline
d8c0755 (HEAD -> collab, origin/collab) Add User 2 fun fact
b60372c Add user1 fun fact
32b6056 (origin/main, origin/HEAD, main) Merge pull request #3 from mareeh58/revert-2-feature/user-profile
7e19adc Revert "Add user profile feature "
ea3f889 Merge pull request #2 from mareeh58/feature/user-profile
8e49e8d (origin/feature/user-profile) Fix: Add error handling as requested in review
3239fe8 add profile feature initial code
1145d58 (origin/staging, origin/develop, staging, feature/new-login, develop) Add feature new login
73ddc7f Merge pull request #1 from mareeh58/feature/db-connection
e85b8f4 (origin/feature/db-connection, feature/db-connection) Add database Configuration
cd0529b Merge branch 'feature-1'
940e115 Add new function to main.py
3af862f Adding notes.txt
07c0640 (origin/bugfix/user-auth-error, feature/db-connection, bugfix/user-auth-error) Initial commit
mareeh58@ubuntu:~/DevOps_Lab2_practice$
```

collab_commit

```
mareeh58@ubuntu:~/DevOps_Lab2_practice$ git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
mareeh58@ubuntu:~/DevOps_Lab2_practice$ git merge collab
Updating 32b6056..d8c0755
Fast-forward
 notes.txt | 2 ++
 1 file changed, 2 insertions(+)
mareeh58@ubuntu:~/DevOps_Lab2_practice$ git push origin main
fatal: 'origin' does not appear to be a git repository
fatal: Could not read from remote repository.

Please make sure you have the correct access rights
and the repository exists.
mareeh58@ubuntu:~/DevOps_Lab2_practice$ git push origin main
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:mareeh58/DevOps_Lab2_Practice.git
 32b6056..d8c0755  main -> main
mareeh58@ubuntu:~/DevOps_Lab2_practice$
```

collab_merge

Summary of Results & Key Learnings

1. Branch Isolation: Developing on isolated feature branches ensures that unfinished code does not affect production.
2. Enterprise Flow: Implementing develop -> staging -> main models mirrors real-world deployment testing pipelines.
3. Code Review Culture: PR reviews, inline comments, and commit iterations ensure high code quality before merging.
4. Clean Repository Hygiene: Deleting merged branches locally and remotely keeps the project repository clean and maintainable.
5. Team Collaboration: Simulated multi-author contributions on shared branches demonstrate real agile team workflows.